

Day -2 : TCP/IP Theory

1. Capturing Flipkart Traffic in Wireshark :

When you open Wireshark and click "Refresh" on Flipkart, your computer sends and receives a lot of tiny data packages (packets). If you click on any single packet in Wireshark and look at the bottom section, you will see 4 main layers attached to it like layers of an onion:

- **Application Layer:** Look for **HTTP** or **TLS/HTTPS**. This is the actual web page content—like the pictures of products, prices, or product titles you see on Flipkart.
- **Transport Layer:** Look for **TCP**. This acts like the delivery manager. Inside, you will see port numbers (like Port 443) that make sure the data goes to the right browser tab on your computer.
- **Internet Layer:** Look for **IPv4**. This acts like the address on an envelope. It shows your computer's IP address (where the request came from) and Flipkart's IP address (where it's going).
- **Network Access Layer:** Look for **Ethernet II**. This acts like the hardware address (MAC address). It shows the digital ID of your laptop and your home Wi-Fi router.

2. OSI vs. TCP/IP Layer Comparison

OSI Model Layer	TCP/IP Model Layer	Real-World App Example
7. Application	Application	Typing a message and hitting send in WhatsApp .
6. Presentation	Application	WhatsApp encrypting your text into scrambled code before sending.

OSI Model Layer	TCP/IP Model Layer	Real-World App Example
5. Session	Application	Staying logged into Zomato so you don't re-enter your password every time you refresh.
4. Transport	Transport	TCP splitting a large picture sent on WhatsApp into numbered chunks and reassembling them on the receiver's end.
3. Network	Internet	Google Maps addressing packets with your IP and the destination server IP to route data across the internet.
2. Data Link	Network Access	Your phone sending Wi-Fi signals containing data frames directly to your home router.
1. Physical	Network Access	Fiber-optic light pulses or electrical signals physically moving through ethernet cables or radio waves.

3. TCP Segment Header Fields Explained (In Simple Terms) :

Think of a TCP header as a **shipping label** attached to a package of data traveling over the internet. Here is what each box on the label does:

- **Source Port:** The return address on your device. It tells the network which specific app or browser tab sent the data.
- **Destination Port:** The receiving address on the server. It tells the other computer which service (like HTTPS web traffic on Port 443) should open the data.
- **Sequence Number:** A tracking number for each piece of data. It helps the receiving phone or computer put scattered data chunks back in the right order.
- **Acknowledgment Number:** A receipt confirmation. It tells the sender, *"I received everything up to piece #500, now send piece #501."*

